PRT362S
Individual Staggered Submission 1

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Content:

Business Rules

ERD diagram of Billing and Payment sections for

Forge Fitness (Gym Website)

Business Rules:

FORGE FITNESS

Business Rules

1.User Authentication and Registration :

1.1 A new member should provide their ID for verification and create a password.

1.2 Password must meet the systems security requirements (min. length characters).

1.3 A user may only get access to member-only features when their membership is active and verified.

1.4 The system should verify login credentials before giving access.

1.5 User are only allowed to edit their own profile they are logged in with.

2.Membership :

2.1 User must be a registered and verified member of the gym to access membership.

2.2 Memberships are provided in various ways depending on the type of member.(Student , normal citizen, or elderly).

2.3 Memberships automatically expire at the end of the package duration.

2.4 Members may upgrade their package at any time, however downgrading is only permitted at the end of the current billing cycle.

2.5 A member may terminate their membership through their profile settings ; their data will be archived into the database and access will be revoked.

3.Packages :

3.1 Only verified and registered members can buy packages.

3.2 Each package must have a defined price , description and duration .

3.3 A member can only have one package at a time (active package).

4.Courses and Videos :

4.1 Course content should be evaluated before being published.

4.2 Each course will include min one video.

4.3 Videos will be provided for explanation purposes only.

4.4 Videos will be provided only to active members.

5. Personal Trainers Information :

5.1 Trainer profile should be provided with a name , qualification and background.

5.2 Only registered trainers may edit their own profile.

5.3 Trainers profiles are visible to all users. (registered or not registered)

5.4 Each trainer has a maximum capacity of assigned members to ensure quality of service.

5.5 Trainers and members can communicate only through the system's internal messaging feature to ensure privacy and record – keeping.

6. Payments and future billings :

6.1 Users pay a registration fee when signing up.

6.2 All package purchases must be recorded in the billing system.

6.3 Users should receive a email when registration was successful.

6.4 Admin may generate and review billing reports.

7. Admin Controls :

7.1 Only administrators are allowed to add and update or even delete members accounts.

7.2 Only administrators may update packages , courses or trainers details.

7.3 Administrators have enhanced privileges on their accounts.

7.4 All changes made by administrators must be logged for audit purposes.

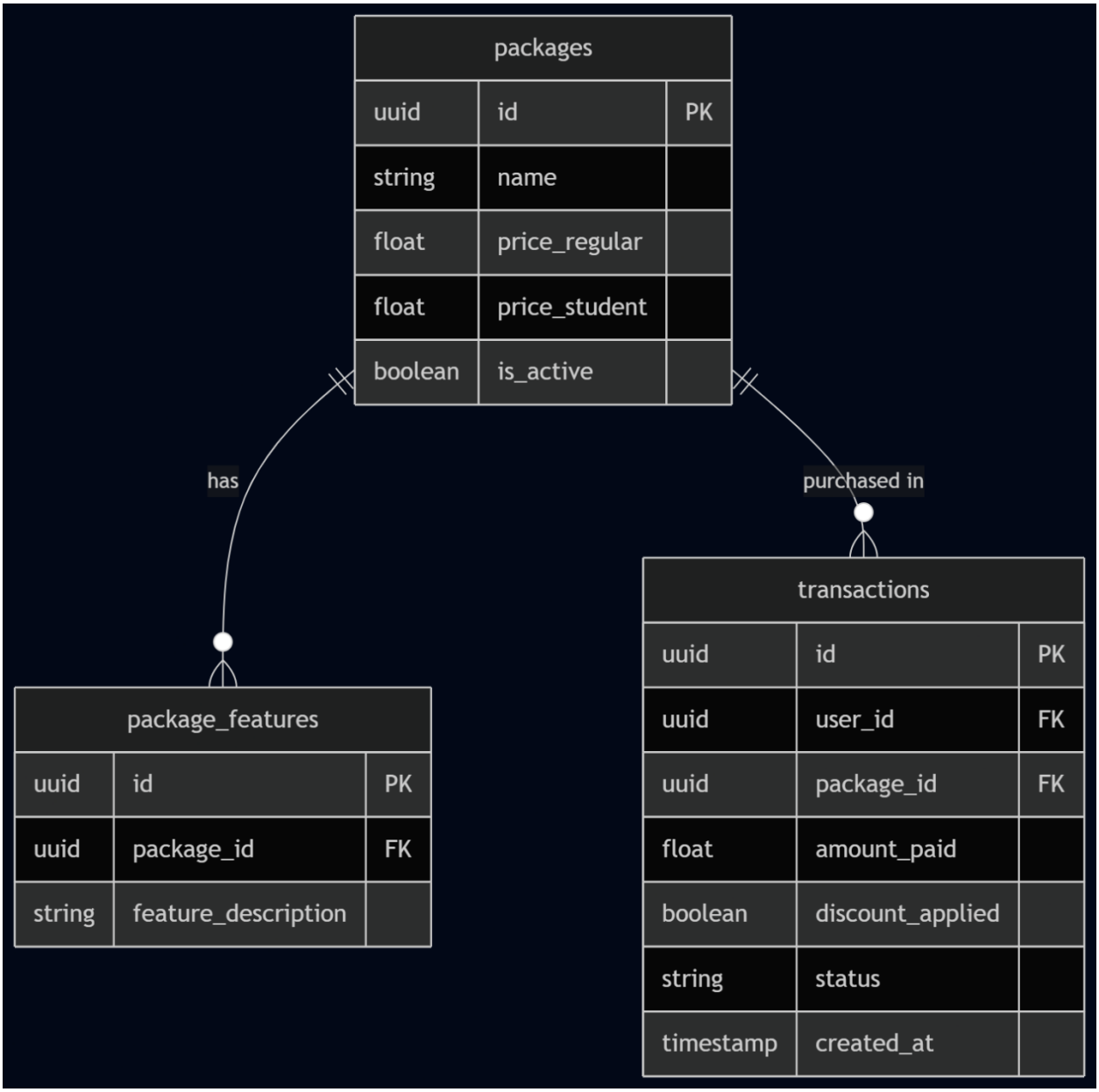
8. System Management

8.1 The System must store all user and membership information in the database.

8.2 The System must back up data regular to prevent risk of losing all data.

ERD Diagram:

PACKAGES AND BILLING



The Packages and Billing module is responsible for managing the membership tiers available on the site and recording all payment activity made by users. The “packages” table stores each membership option the gym offers, holding both a regular price and a discounted student price, as well as an active flag that allows the gym to enable or disable a package without deleting it.

The “package_features” table stores the individual features listed under each package as displayed on the Packages page, such as course access levels or trainer session allowances. When a user selects a package and completes payment, a record is created in the “transactions” table which logs the user, the package purchased, the amount paid, whether a student discount was applied, and the status of the payment. This module connects directly to the Auth module since a user must be logged in and their student verification status must be checked before a discount can be applied. It also indirectly governs access to the Courses and Training Plans modules since the package a user holds determines what content they are able to unlock on the site.

The Billing page handles the payment interaction itself while the Packages page handles the selection and display of available tiers.